§5 Standard operation form and inter-time connection — part 1 of 2

Reference: sec:extended_dynamics, subsec:update_dependency, subsec:s5_operation, subsec:memory_dyn, subsec:s5_continuation, subsec:s5_record_handover, subsec:s5_observation

Source: paper/section5_extended_dynamics.tex, lines 1–293 (responsibilities R1–R4; the standard step; learning and working components and retained records; one-step well-formedness and continuation of the operation record; handover of retained records; observation factorization of Update), at commit e6e12cd. Status: draft.

Opening and responsibilities

Reference: sec:extended_dynamics, subsec:update_dependency, subsubsec:update_role

§5 connects the types and the update of §2, the description level of §3 and the same-time structure of §4. The object is a deterministic, discrete-time, single ACS. Once the standard maps of the primitive functions and the readout rules are fixed and an environment realization (def:env_realization) is given, a one-step formal record is determined and closes to the next time. The existence of a mathematical record is distinguished from the activity, well-formed realization and persistence of an actual individual.

Definition (def:s5_responsibilities): responsibilities of this section. R1: give a type-consistent one-step operation record including the behavior-responsive component. R2: connect state, internal model and behavior-responsive component to the next time using the existing f, Update and the g_O of the environment realization. R3: for a fixed sequence of background components, initial values and standard maps, show existence and uniqueness of the mathematical record. R4: hand the feasible action set, the operation record and the classification-C quantities to later sections. R3 does not mean maintenance of feasibility, non-blockage, non-collapse or continued activity of the implementation.

Boundary of responsibilities (rem:s5_scope_boundary). §5 does not redefine the responsibilities of mechanism, base and implementation of §4. Relational description follows §6; termination, collapse and the interpretation of their records follow §2 and §7. Evaluating a total map after functional termination is not a claim that the implementation operates after it.

The standard step

Reference: subsec:s5_operation

Definition (def:s5_standard_step): one step of standard operation. Fix the total single-valued maps of def:three_primitives and the environment realization $(E^{\mathrm{bg}}, O, g_O, (b_t)_{t\geq 0})$. For $(x_t, M_t, o_t) \in X \times \mathcal{M} \times O$ and $b_t \in E^{\mathrm{bg}}$: $e_t = (b_t, o_t)$, $I_t = P_{\mathrm{sens}}(x_t, e_t)$, $r_t = P_{\mathrm{disc}}(M_t, I_t)$, $\nu_t = P_{\mathrm{val}}(M_t, r_t)$, $u_t = P_{\mathrm{act}}(M_t, r_t, \nu_t)$, $x_{t+1} = f(x_t, u_t, e_t)$, $M_{t+1} = \mathrm{Update}(M_t, x_t, e_t)$, $o_{t+1} = g_O(o_t, x_t, u_t, b_t)$. The action u_t includes the resolution by PSL and is not identified with the M3 command; a concrete form follows def:s4_primitive_composition. The values x_t, M_t, o_t are those at the start of the step, and no same-time output is fed back as input. The mechanism outputs of §4 (the purpose set Pur_t, the purpose status judgment stat_t and others) are generated from (M_t, I_t) under fixed rules (def:s4_carrier_contract), so they are determined by the values of the step and are not added as separate components.

Definition (def:update_section5, def:s5_step_map): reference to Update and the one-step transition map. Using Update : $\mathcal{M} \times X \times E_{\mathrm{ext}} \to \mathcal{M}$ of §2 and g_O, the one-step transition map $\Phi_{\mathrm{step}} : X \times \mathcal{M} \times O \times E^{\mathrm{bg}} \to X \times \mathcal{M} \times O$ is $\Phi_{\mathrm{step}}(x, M, o, b) = (f(x, u, e), \mathrm{Update}(M, x, e), g_O(o, x, u, b))$, with $e = (b, o)$ and r, ν, u computed as in the standard step. The product $X \times \mathcal{M} \times O$ is a product of existing state spaces, not an eighth tuple component; g_O is not a component of the 7-tuple but a structure on the side of the environment realization. Under ass:compact_spaces and ass:M_compact, $X \times \mathcal{M}$ is compact. Compactness of O is not assumed (def:Eext_decomposition), so compactness of the whole product is not stated. §5 neither assumes nor concludes continuity of Φ_{step}. When $\mathcal{A}_O = \emptyset$, O is a single point and

the third component is constant.

Learning and working components, retained records

Reference: subsec:memory_dyn, subsubsec:s5_memory_update

Definition (def:memory_intertime_update): inter-time readout of retained content. With $M_t^{\mathrm{L}}, M_t^{\mathrm{W}}$ of def:model_learning_working and the fixed readouts of def:fast_memory_sets, $\mathcal{T}_{\ell,t} = \rho_\ell(M_t^{\mathrm{L}})$ and $\mathcal{T}_{\ell,t+1} = \rho_\ell(\mathrm{pr}_L\,\mathrm{Update}(M_t, x_t, e_t))$. Empty values are allowed, and no independent update map is placed on retained content. ρ_ℓ is a total map, but whether the retained content is actually usable is a separate question.

Carrier of M1 internal state and history (rem:s5_m1_state_ref). $\mathrm{S}_{M1,t}$ is a bundle of same-time outputs; later reference uses the retained record of def:s4_carrier_contract. This paper does not state as a general consequence that the product of two components or the type of a comprehensive Update retains arbitrary history exactly, or that arbitrary information can be written between mechanism components. The dependency order of component updates is distinguished from the generation dependency of M1, M2 and M3 at the same time.

One-step well-formedness and continuation

Reference: subsec:s5_continuation

Proposition (prop:s5_one_step): P1, one-step well-formedness. Fix the total single-valued maps of def:three_primitives, def:state_transition, def:update_operator and g_O of def:env_realization. Then for each (x, M, o, b) every value of def:s5_standard_step and $\Phi_{\mathrm{step}}(x, M, o, b)$ are uniquely determined, and the latter belongs to $X \times \mathcal{M} \times O$. The proof follows the types in order and uses single-valuedness; no downstream output of the same time is required as an input.

Definition (def:s5_operation_record): standard operation record. Given an environment realization and an initial value (x_0, M_0, o_0), the sequence $(x_t, M_t, o_t, b_t, e_t, I_t, r_t, \nu_t, u_t)_{t\geq 0}$ following def:s5_standard_step is called a standard operation record. Only the sequence of background components is given exogenously; o_t can respond to the past actions of the ACS (rem:exogenous_environment). The set of state–environment subsequences $(x_t, M_t, e_t)_{t\geq 0}$ is written $\mathcal{Z}_{\mathrm{op}} \subseteq (X \times \mathcal{M} \times E_{\mathrm{ext}})^{\mathbb{N}_0}$ and is relative to the fixed standard maps and g_O. Closedness and compactness are not required. When $\mathcal{A}_O = \emptyset$ the definition coincides with a record given by regarding the background sequence as the external-state sequence.

Proposition (prop:s5_continuation): P2, continuation of the mathematical operation record. Under the conditions of P1, for any background sequence and initial value there is exactly one standard operation record on all $t \in \mathbb{N}_0$ (recursion on the natural numbers for existence, induction for uniqueness).

Proposition (prop:s5_causality): P3, causality of the operation record. Under fixed standard maps and g_O, (x_t, M_t, o_t) is determined by (x_0, M_0, o_0) and $b_0, \ldots, b_{t-1}$ alone; the same-time $e_t, I_t, r_t, \nu_t, u_t$ additionally use b_t. Since o_t can depend on past actions, causality is stated with respect to the background sequence.

Formal continuation and continuation of realization (rem:s5_formal_not_survival). P1–P3 do not use $K, A_t, \mathrm{WF}_{i,t}, \mathrm{FT}_{i,t}, \mathrm{Col}_{i,t}$ as premises. Since P_{act} is total, a formal action value u_t is determined even at times with $A_t = \emptyset$, but this value is not taken to be a feasible action; the record also continues from $x_t \notin K$. The treatment of mechanism outputs at times with $\mathrm{WF}_{i,t} = 0$ follows def:s4_output_attribution. A formal record after functional termination or collapse is not read as the realization of the activity of the same individual. The description domain of the same ACS follows $I_{\mathrm{ACS}}(\sigma)$ of §7, and principle judgment further follows $\mathbb{T}_{\mathrm{P},i}$ and the domain of FR of §3.

Handover of retained records

Reference: subsec:s5_record_handover

Definition (def:s5_record_agreement): agreement of retained records (classification C). Fix a standard operation record and the readout ρ_{past} of def:s4_carrier_contract. For $k = 1, \dots, 7$ the corresponding value $y_{k,t}$ is: (1) Priority P_t; (2) effective action u_t; (3) the sequence of state estimates $\hat{x}_s$ from $\max\{0, t-W+1\}$ to t; (4) the bundle $\mathsf{S}_{M1,t}$ of M1 outputs; (5) the sequence of sensed information I_s from $\max\{0, t-W_{\mathrm{short}}+1\}$ to t; (6) the purpose set Pur_t; (7) the outside-side prediction set $\hat{G}_t(u_t)$ for the effective action (def:env_set_prediction). If $\mathrm{pr}_k(\rho_{\mathrm{past}}(M^{\mathrm{W}}_{t+1})) = y_{k,t}$, the k-th component of the record *agrees* at time t. Other retained-record readouts, such as $\rho_{\mathrm{eval,past}}$ of def:m2_latent_instability, are defined in the same form by specifying the corresponding past values.

Whether the record agrees depends on the detailed type of Update and the content of the working component. This paper neither states the agreement as a general consequence nor places it as an assumption (def:s4_carrier_contract). prop:s5_continuation does not presuppose agreement. Since the mechanisms of §4 are total maps returning values whatever the record is, the operation record is uniquely determined including times at which the record does not agree.

Handover of purposes (rem:s5_purpose_handover). The previous-time purpose set in the input of M3 at $t+1$ is the record value of the 6th component. The domain of the purpose status judgment stat_{t+1} is, in the standard form, this record value (def:purpose_status), and equals Pur_t only if the 6th-component record agrees at t. When it does not agree, the course of purposes is described within the range of def:purpose_course. A purpose missing from the record can leave the purpose set without a judgment. A purpose appearing only in the record and taken as continuing by M3 is not in Pur_t, so in def:purpose_course it is treated as newly entering the purpose set at $t+1$.

Persistence of internal object tokens (rem:s5_token_persistence). Persistence of tokens across times (rem:token_generation) is expressed as M1 at $t+1$ using the 4th-component record value as one input and giving the same label as at t. Whether it gives the same label is determined as a value of the M1 map; §5 places no separate persistence rule. Token persistence here is therefore the inter-time continuation of an internal label. Continuation of a label, $k \in \mathcal{K}_t \cap \mathcal{K}_{t+1}$, is determined on the operation record as a comparison of the M1 outputs at two times. Whether the same label points to the same external frame $a \in \mathcal{A}_O$ or the same individual is a separate condition that can be asked once a describer supplies a correspondence between labels and the outside; this paper does not define that correspondence.

Observation factorization of Update

Reference: subsec:s5_observation

Assumption (ass:s5_observation_factorization): observation factorization of Update. There is a total map $\Phi_{\mathrm{upd}} : \mathcal{M} \times \mathcal{J} \to \mathcal{M}$ with $\mathrm{Update}(M, x, e) = \Phi_{\mathrm{upd}}(M, P_{\mathrm{sens}}(x, e))$ for all (M, x, e). Introduction status: S (structural). Strength: medium. It is an additional condition excluding systems in which state or environment differences other than observation directly affect the update, and it is not made a constitutive requirement of an ACS in §2. Continuity of Φ_{upd} is not assumed. It is relied on in prop:s5_observation_causality; P1–P3 do not need it.

Proposition (prop:s5_observation_causality): check condition for observation factorization, and causality of the internal model. The assumption is equivalent to: for each fixed M, $P_{\mathrm{sens}}(x, e) = P_{\mathrm{sens}}(x', e')$ implies $\mathrm{Update}(M, x, e) = \mathrm{Update}(M, x', e')$. Under it, M_t is determined by M_0 and $I_0, \dots, I_{t-1}$ alone. (The converse extends Φ_{upd} outside $\mathrm{Im}\, P_{\mathrm{sens}}$ by a fixed element of $\mathcal{M} \neq \emptyset$; at $t = 0$ the input history is empty and M_0 itself is needed.)

Scope of the factorization (rem:s5_observation_scope). Observation factorization is not equivalent to knowing the true value. A realization in which an unobserved change of the base acts on the internal model may fail the assumption. The existence of Φ_{upd} alone does not imply that the detailed types Update_1–Update_3 of §2 can be realized. §5 treats updates satisfying both the existing component decomposition and the factorization. Φ_{upd} is a representation of the update component and does not uniquely derive its structural role from

sensing. Non-redundancy of the 7-tuple is maintained in the sense of the role separation of `rem:tuple_nonredundancy`.

Deviation from prediction and update of the internal model (`rem:s5_prediction_learning`). The theoretical deviation from prediction of `def:prediction_record_check` is a quantity checking against the realized value o_{t+1}. Under the assumption, by `prop:s5_observation_causality`, o_{t+1} enters the internal model only through the sequence of sensed information. §5 gives no update rule based on deviation from prediction (`rem:prediction_reference_difference`). The prediction accuracy coefficient $\pi_{\tau,t}$ (`def:s5_prediction_coefficient`) is a total coefficient specified by the describer and takes a value even when the prediction set is $\perp$. Each of the two deviations is undefined when its own reference prediction set is $\perp$; one being undefined does not imply that the other is undefined. Their additional well-formedness conditions ($\mathrm{WF}_{i,t+1} = 1$ for the internal one, $\mathrm{WF}_{i,t} = 1$ for the theoretical one) follow `def:prediction_record_check`. This paper does not identify the two deviations or state a general relation between them.

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.